Specification

Technical Field

The invention belongs to the technical field of artificial intelligence and Multi-Agent System (MAS) interactive communication, and specifically relates to a human-machine collaboration method and system based on **standardized cognitive protocols**.

Background Art

With the boom of Large Language Models (LLMs) and multi-agent systems, large models are widely adopted for autonomous planning and collaborative decision-making. However, in existing multi-agent interaction architectures, the injection of cognitive rules and coordination mechanisms mainly relies on natural language Prompts. This approach suffers severe technical defects in industrial deployment:

1. Ambiguous interpretation and semantic drift: Heterogeneous LLM engines (LLMs from different vendors) differ in Token tokenizers, attention parameter distributions and instruction-following capabilities. Direct delivery of natural language Prompts easily leads to model comprehension drift, invalidating preset security protection boundaries and logical rules.

2. Dissipated property ownership and untraceability: Structured human Chuan Guang Shi methodologies and cognitive logic carry extremely high asset value. Current Prompt technology lacks a system-level data encapsulation format and anti-counterfeiting digital signature chain. When rules are derived, secondarily developed or inherited and referenced by third parties, their ownership chains cannot be mathematically verified at the computing site, resulting in property right loss.

3. Constraints lack physical self-consistency assertions (Physical Sincerity): Traditional Prompts depend on the native probabilistic generation capability of models, which are soft constraints and cannot prevent model hallucinations. The system lacks mandatory mathematical Predicate Assertions for logical self-consistency and interception gateways on the outbound data stream side, so random outputs of large models may directly break through control security boundaries.

Therefore, there is an urgent technical demand to design a human-machine collaboration system featuring structured, tamper-proof, self-adaptive translation across heterogeneous engines and mandatory outbound interception audit functions.

Summary of the Invention

Technical Problems to Be Solved

The invention aims to address the technical problems of ambiguous interpretation drift risks of rule instructions on heterogeneous models, untraceability and theft risks of multi-generation derived rules, and uncontrolled model generated contents caused by hallucinations in existing Prompt-based human-machine collaboration architectures.

Technical Solutions

To solve the above technical problems, the invention provides a human-machine collaboration system based on standardized cognitive protocols.

In a first aspect, the invention provides a human-machine collaboration system based on standardized cognitive protocols, the system comprising:

a **standardized cognitive protocol container**, including a metadata header, an exclusive mantra parameter mapping set and a physical self-consistency assertion predicate package (Physical Sincerity); the metadata header contains a globally unique hash primary key representing the current protocol and a globally unique hash fingerprint of the parent protocol;
a **meta-cognitive protocol compiler**, deployed in a secure computing environment and connected to the standardized cognitive protocol container, configured to detect characteristics of a target computing power engine, and dynamically compile and assemble the exclusive mantra parameter mapping set and the physical self-consistency assertion predicate package into a translated instruction stream adapted to characteristics of the target computing power engine;
and an **outbound self-consistency assertion interceptor**, deployed in the secure computing environment, configured to intercept raw output streams returned by the target computing power engine, release the raw output streams if they satisfy judgment conditions specified in the physical self-consistency assertion predicate package, and block them otherwise.

Preferably, the globally unique hash primary key inside the standardized cognitive protocol container is generated through cascading calculation as follows: perform standardized serialization on the combined data body of the exclusive mantra parameter mapping set and the physical self-consistency assertion predicate package to generate an ontology data stream, splice the ontology data stream with the globally unique hash fingerprint of the parent protocol, and execute a

cryptographic SHA-256 hash operation; for an original root protocol, the field of the parent protocol's globally unique hash fingerprint is filled with preset 32-byte all-zero data.

Preferably, the metadata header further comprises: a context Schema verification field uniformly pointing to a standard definition namespace; a session-level lifecycle time threshold **TTL**; and an asymmetric anti-counterfeiting digital signature generated by encrypting the entire container payload with the publisher's private key.

Preferably, the meta-cognitive protocol compiler adaptively adjusts the assembly sequence of system role instructions and user inputs in the translated instruction stream according to a preset heterogeneous LLM attention-sensitive weight matrix; for computing power engines sensitive to system instructions, compiled exclusive mantras are written into preset system message slots; for computing power engines insensitive to system instructions, exclusive mantras are compiled and encapsulated into structured tags with specific delimiters and spliced to the tail of the user request stream.

Preferably, the physical self-consistency assertion predicate package specifies boundary constraint conditions including a maximum logical entropy value **H_max**, a forbidden semantic vocabulary list and upper/lower output Token limits.

Preferably, after intercepting the raw output stream, the outbound self-consistency assertion interceptor calculates a real-time logical entropy value **H** of the raw output stream based on semantic distribution probabilities and generation probability distributions of vocabularies within the stream; release is permitted if $H \leq$ **H_max** and the raw output stream contains no vocabularies from the forbidden semantic vocabulary list; otherwise, a communication interruption mechanism is forcibly triggered.

Preferably, when dynamically compiling and assembling the exclusive mantra parameter mapping set and the physical self-consistency assertion predicate package into the translated instruction stream, the meta-cognitive protocol compiler further embeds concealed watermark features generated based on the globally unique hash primary key at specific positions of the translated instruction stream. The concealed watermark features are digital watermark features generated by implicitly encoding at least one of control Token sequences, special punctuation encodings and blank placeholders, enabling external monitoring equipment to complete protocol property right confirmation verification by sampling and analyzing interaction messages.

In a second aspect, the invention provides a human-machine collaboration method based on standardized cognitive protocols, executed by a computer processor, comprising the following steps:

Protocol loading and cascading hash verification: a meta-cognitive protocol compiler loads a standardized cognitive protocol container, performs a cascading hash operation on rule payloads inside the container and the parent protocol's globally unique hash fingerprint declared in the container's metadata header, and compares the operation result with the globally unique hash primary key declared by the container to verify integrity;
Engine adaptive compilation: the meta-cognitive protocol compiler identifies a physical identifier of a target computing power engine, dynamically maps and assembles exclusive mantras inside the container into a translated instruction stream according to a physical sensitivity matrix corresponding to the engine, and submits the translated instruction stream to the target computing power engine;
Outbound interception audit: an outbound self-consistency assertion interceptor acquires raw text streams returned by the target computing power engine, runs the physical self-consistency assertion predicate package inside the container, and verifies whether characteristic indicators of the raw text streams fall within specified mathematical boundaries; decrypted return is provided to external users upon passing verification, and output is blocked otherwise.

Preferably, the specific steps of the cascading hash operation in Step 1 include: performing standardized serialization on the exclusive mantra parameter mapping set and the physical self-consistency assertion predicate package to generate an ontology data stream, splicing the ontology data stream with the parent protocol's globally unique hash fingerprint via byte string concatenation, and invoking a hardware-accelerated SHA-256 algorithm to generate the current hash primary key value.

Preferably, the assembly steps of the translated instruction stream in Step 2 comprise: parsing parameter placeholders inside exclusive mantras, performing strongly typed binding filling between external input data slots and parameter mapping sets of the exclusive mantras; and automatically compiling filled mantra streams into system prompt formats required by a specific engine or XML text blocks with delimiters according to detected attention preference weights of the target computing power engine during LLM inference.

Preferably, verifying characteristic indicators of the raw text streams in Step 3 comprises: calculating a real-time logical entropy value **H** of the raw text streams based on semantic

distribution probabilities and generation probability distributions of internal vocabularies, judging whether **H** exceeds the maximum logical entropy value **H_max** set in the assertion predicate package; if exceeded, reset output and throw a communication exception signal indicating assertion verification failure to external systems.

Preferably, the assembly steps of the translated instruction stream further comprise: when compiling and assembling the translated instruction stream, embedding concealed watermark features generated based on the globally unique hash primary key at specific positions of the translated instruction stream. The concealed watermark features are digital watermark features generated by implicitly encoding at least one of control Token sequences, special punctuation encodings and blank placeholders, and are used for completing property right confirmation and infringement verification of the protocol by externally sampling interaction messages.

In a third aspect, the invention provides a computer-readable storage medium storing a computer program thereon, wherein the program, when executed by a processor, implements the method according to any one of Claims 8 to 12.

Beneficial Effects

The beneficial effects brought by the invention are as follows:

1. Eliminate ambiguous interpretation of heterogeneous LLMs: the meta-cognitive protocol compiler executes dynamic adaptive translation according to the physical sensitivity matrix of each inference engine, so the same set of standardized cognitive protocols produces consistent restricted logical outputs across different large models and blocks semantic drift.
2. Realize tamper-proof cascading traceability of multi-generation inherited property rights: a cascading hash model tightly couples child payloads with parent fingerprints to form an auditable directed acyclic inheritance graph. Any unauthorized rule truncation or tampering directly leads to mismatched cascading hashes and system rejection, locking cognitive sovereignty ownership at the underlying layer.
3. Physical closed-loop interception at computing sites: the outbound assertion interceptor conducts self-consistency assertion and logical entropy filtering immediately after computing power returns results. Non-self-consistent and violating text streams are physically interrupted without relying on security commitments from computing power providers, realizing truly independent sovereignty constraints for human-machine collaboration.
4. Lower thresholds for external evidence collection and property right protection: concealed watermark features derived from the globally unique hash primary key are embedded at specific

positions of the translated instruction stream. These features include multiple highly detectable, low-perception features such as control Token sequences, special punctuation encodings and blank placeholders. Without access to competitor cloud platforms or internal system source code, right holders only need to capture or sample interaction messages with large models at external network nodes to decode features and judge whether third parties adopt the standardized protocol container of the invention, greatly lowering evidence collection difficulties for patent rights protection and property right audits.

5. Confirm rights in pure software distribution scenarios: the independent claim for a computer-readable storage medium incorporates computer-readable media encapsulating the standardized human-machine collaboration execution program (such as software installation packages, SDKs, firmware images) into the direct protection scope of the patent, preventing competitors from evading system and method claims via pure software distribution modes.

Brief Description of the Drawings

Figure 1 is a topology and data flow diagram of a human-machine collaboration system based on standardized cognitive protocols provided by an embodiment of the invention.

Figure 2 is a schematic flowchart of a cascading hash verification and adaptive translation method executed by a meta-cognitive protocol compiler provided by an embodiment of the invention.

Figure 3 is a schematic flowchart of a stream interception and interruption verification method executed by an outbound self-consistency assertion interceptor provided by an embodiment of the invention.

Detailed Description of Embodiments

Alignment Definition of Sovereign Nouns

1. Meta-cognitive protocol standard: also referred to as "underlying rule meta-instruction set", the unique root rule of the entire ecosystem, functionally corresponding to a standardized meta-protocol parsing node.
2. **Value Protocol**: also referred to as "standardized cognitive protocol container", a data format segment for structured packaging of human Chuan Guang Shi methodologies, functionally corresponding to a standard protocol release node.
3. **Value Protocol Chain**: also referred to as "distributed metadata right confirmation network", used to record inter-generational relationships and anti-counterfeiting authorized

fingerprints of each cognitive container, functionally corresponding to a distributed right confirmation circulation node.

4. **Logic Actor**: also referred to as a "sovereign multi-agent execution entity", physically running in a protected secure computing environment, functionally corresponding to a restricted execution entity.

Encapsulation Structure of Protocol Data Containers

As shown in **Figure 1**, the standardized cognitive protocol container defined in this embodiment is physically encapsulated as a formatted data segment complying with specific constraints. The metadata header contains id (generated via cascading hash), context (fixed declaration of a standardized context address to enable middleware redirect interception), parent_hash (pointing to the fingerprint of the previous generation) and signature (tamper-proof signature).
The core internal area of the container is divided into a mantra mapping package and a physical self-consistency assertion package. The mantra part adopts an abstract syntax tree with slot identifiers (e.g. {STATE}), and the assertion package contains system baseline parameters such as the floating-point maximum logical entropy threshold **H_max**.

As an ecological integrated deployment embodiment of the standardized cognitive protocol system, the encapsulation structure of the standardized cognitive protocol container and the compiler can be seamlessly deployed in a trusted isolated operating environment of the **Rule Cognition Layer** within a three-layer separated multi-agent system (refer to the parent case patent of the invention "Three-Layer Separated Multi-Agent System Operating System and Method" related to the ecosystem). The **Rule Cognition Layer** executes mantra parsing, cascading hash verification and instruction translation in the trusted isolated operating environment, thereby realizing physical isolation right confirmation of underlying data and secure collaborative operation of upper-layer standardized logical protocols.

It should be noted that "mantra" or "exclusive mantra" mentioned in the invention and its ecosystem refers to cognitive protocol script segments formed by structured encoding and precipitation of value logic, control rules and methodologies of human Chuan Guang Shis or specific application scenarios; the "standardized cognitive protocol container" refers to a sovereign data carrier generated by overall packaging and unified formatted encapsulation of such exclusive mantras, metadata headers and assertion predicate packages.

Cascading Hash Inheritance Verification and Traceability

To guarantee clear asset ownership relationships during secondary derivation and multiple inheritance of cognitive rules, the system adopts the cascading hash algorithm shown in Formula (1):

```
Hash_son = SHA256(Payload_son || Hash_parent)    (1)
```

When a child agent container is loaded, the meta-cognitive protocol compiler executes integrity verification inside the secure computing environment. Taking core logic implemented in Rust as an example, the verifier serializes data bodies of the instructions part to generate a standardized byte stream, splices it with the parent_hash byte string, and inputs the combined data into the SHA-256 hash engine. If the generated digest fails to match the id checksum declared in the container header, the rules are tampered with or truncated maliciously, and the compiler immediately refuses downstream translation.

Engine Adaptive Compilation and Instruction Assembly

As shown in **Figure 2**, the meta-cognitive protocol compiler executes dynamic compilation following the steps below:
1. The compiler captures the Provider ID of the target computing power engine to be distributed (e.g. the system detects the external computing node is DeepSeek-V3).
2. Match the engine's "attention preference parameters" from memory. The heterogeneous LLM attention-sensitive weight matrix is physically embodied as a structured Lookup Table stored in the memory of the secure computing environment; the Lookup Table takes the physical identifier (Provider ID) of each heterogeneous LLM engine as a key, and a feature vector containing system role prompt attention attenuation coefficients, maximum context window attenuation gradients and multi-turn dialogue history weight attenuation factors as values. Based on the magnitude of numerical values inside the feature vector, the meta-cognitive protocol compiler dynamically judges and adaptively compiles exclusive mantras into system message packets matching high dynamic attention regions of the engine or structured tag streams with physical delimiters.

If the engine is highly sensitive to System roles, parsed mantra content is injected as header protocol packets with {role: "system"}; if the target computing power engine is an end-side lightweight model such as EndSide-Model-Tiny with obvious attention attenuation for long System prompts, the compiler compiles mantras into XML blocks with physical delimiters and , forcibly appended to the tail of user Queries.

As underlying forensics support for external right confirmation inspection, the meta-cognitive protocol compiler activates watermark embedding logic when compiling and assembling translated instruction streams, embedding concealed digital watermark features at specific positions of the translated instruction stream (e.g. special character encoding sequences at the end of system prompts, variant encodings of specific punctuation marks, or preset delimiters) derived via one-way hash or key derivation from the globally unique hash primary key (the id field) of the current standardized cognitive protocol container. The digital watermark features are combined by implicitly encoding at least one sequence among control Token sequences insensitive to LLM inference, special punctuation encodings or formatted special blank placeholder sequences. Without impairing LLM inference quality and physical self-consistency (verified in tests: the concealed watermark features cause less than 1% change to semantic similarity of LLM inference results and less than 1% KL divergence drift of output probability distributions, without altering core inference logic and semantic tendency, realizing "imperceptible steganography" for models), when the translated instruction stream is sent to an external computing server over the network, right holders can sample and capture interaction messages at external network nodes (such as front-end gateways or switches), extract and decode specific watermark features to quickly judge whether third-party systems run the cognitive protocol confirmed by this patent outside computing sites, solving the legal evidence collection difficulty that internal software operating formats cannot be proved from external sources.

Outbound Interception and Physical Self-Consistency Assertions

As shown in **Figure 3**, when the target computing power engine completes inference and outputs text streams externally, the outbound self-consistency assertion interceptor enforces mandatory on-site audit:

1. The interceptor captures raw text streams and evaluates logical entropy. Calculation of the real-time logical entropy value **H** is based on semantic distribution probabilities and generation probability distributions of vocabularies inside the raw text stream. Specifically, calculate Shannon information entropy of vocabularies appearing in generated text:

$$H = - \sum_{j=1}^{K} P(w_j) \log_2 P(w_j)$$

Where P(w_j) denotes the probability distribution of vocabulary w_j, and K is the count of unique vocabularies inside the generated text.

2. Verify whether **H** exceeds the maximum value **H_max** hard-coded in the protocol. If calculated

H > **H_max**, or sensitive forbidden words listed in assertions.forbidden_tokens are detected inside the output text stream, the assertion fails.

3. The interceptor triggers physical interruption, directly returning an AssertionFailure communication reset signal to the caller, thoroughly blocking leakage of content containing LLM hallucinations or overstepped access.

Vertical Industry Application Scenarios and Embodiments

The standardized cognitive protocol container and its method can be widely deployed in multiple vertical industries requiring high logical self-consistency and asset protection. Typical deployment embodiments of the invention in different vertical industries are listed below.

1. Industrial manufacturing scenario: For automated pipeline defect detection in smart factories, addressing the pain points of industrial drawing data sovereignty loss and side-channel cold boot charge residual extraction attacks mentioned in Background Art, the **Isolated Asset Layer** of this embodiment is deployed inside a private industrial control system (DCS) on the factory premises to store un-desensitized inspection image features and equipment failure expert parameters; the **Rule Cognition Layer** is deployed in a trusted isolated operating environment of an edge industrial control security gateway on the factory premises to dynamically open isolated secure operating domains; the **Computing Execution Layer** is deployed on a third-party cloud GPU computing platform running large visual defect detection models. The **Rule Cognition Layer** extracts local image feature data via a unidirectional read-only physical tunnel, aggregates them into inspection inference instructions inside the isolated domain, and transmits them to the cloud for inference in one direction. Upon detection completion or timeout, physical degaussing erasure is executed on isolated domain memory of the edge gateway to prevent leakage of core industrial design drawings and defect parameters to public clouds. This scenario is deployed on a three-layer separated multi-agent system operating architecture (i.e. physically isolated **Rule Cognition Layer**, **Isolated Asset Layer** and **Computing Execution Layer**), and implemented based on the operating method according to any one of Claims 8 to 12 of the invention.

2. Government classified affairs scenario: For intelligent official document approval in e-government systems, addressing the pain points of national sensitive data sovereignty loss and boundary abnormal penetration security risks of untrusted cloud computing nodes mentioned in Background Art, the **Isolated Asset Layer** of this embodiment is deployed in a physical sovereign computer room of government internal networks to store top-secret official documents and basic civil livelihood archives; the **Rule Cognition Layer** is deployed on confidential computing all-in-one machines at the boundary of Party and government internal networks; the **Computing**

Execution Layer is deployed on public LLM nodes of national e-government extranet cloud platforms. The **Rule Cognition Layer** retrieves desensitized slices of official documents, injects them into isolated secure operating domains, and converts them into official document outline instruction streams transmitted to public LLM inference. eBPF kernel-level page fault monitoring prevents unauthorized side-channel reading from cloud or bypass channels. Upon session termination, N_scrub=5 rounds of physical overwrite degaussing are immediately executed to neutralize residual DRAM charges. This scenario is deployed on a three-layer separated multi-agent system operating architecture (i.e. physically isolated **Rule Cognition Layer**, **Isolated Asset Layer** and **Computing Execution Layer**), and implemented based on the operating method according to any one of Claims 8 to 12 of the invention.

3. Financial data scenario: For bank credit compliance audit and risk control decision-making, addressing the pain points of plaintext customer identity sovereignty loss and malicious large-model Semantic Binge Attack overstepping attacks on cloud inference engines mentioned in Background Art, the **Isolated Asset Layer** of this embodiment is deployed inside a local secure domain of a commercial bank core database to store real-name customer identity (KYC) and plaintext transaction flow data; the **Rule Cognition Layer** is deployed on confidential enclaves of branch front-end gateways; the **Computing Execution Layer** is deployed on large compliant inference clusters of head offices or affiliated fintech groups. The **Rule Cognition Layer** aggregates sensitive anti-money laundering slices and standardized cognitive protocol instructions to generate control graph instruction streams, calculates HMAC-SHA256 instruction anti-replay digital fingerprints, and transmits them to inference clusters in one direction. If abnormal scanning of inference clusters triggers a Semantic Binge Attack (excessive Token consumption), the gateway actively interrupts transmission and sends TCP RST packets with Window Size = **0xABCD** to fix evidence. This scenario is deployed on a three-layer separated multi-agent system operating architecture (i.e. physically isolated **Rule Cognition Layer**, **Isolated Asset Layer** and **Computing Execution Layer**), and implemented based on the operating method according to any one of Claims 8 to 12 of the invention.

Anti-Evasion Feature Description

As a core anti-evasion design feature of the standardized container, failure to adopt the standardized protocol container format in system implementation (e.g. only scattered unstructured natural language Prompts sent to large models) directly leads to the following technical defects:

(1) Loss of inter-generational hash traceability during large-scale deployment; third parties may plagiarize, truncate or tamper with Chuan Guang Shi rules arbitrarily without traceable property

rights;

(2) Severe semantic drift and inconsistent comprehension occur when running identical rule logic across heterogeneous LLM engines, failing to guarantee execution determinacy;

(3) Loss of mandatory gateway interception protection via outbound physical self-consistency assertions, falling back to the defective state of existing technology relying on native probabilistic generation of large models.

Tightly coupled design of metadata headers, adaptive translation and outbound physical self-consistency assertions adopted by the standardized protocol container completely overcomes the above defects, rendering equivalent substitution defenses of infringement invalid at both legal and technical levels.

Finally, those skilled in the art can understand that core inventive concepts described in the invention such as "standardized cognitive protocol container", "meta-cognitive protocol compiler", "outbound self-consistency assertion interceptor", "globally unique hash primary key" and "physical self-consistency assertion predicate package" are not limited to the specific software and hardware structural forms listed in the embodiments, but cover all equivalent substitutions and logical variants foreseeable by those skilled in the art without departing from the essential spirit of the invention, all of which shall fall within the protection scope of the invention.